

# LAKSHYA AGARWAL

## Creative AI Engineer & Full-Stack Systems Architect

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### EXECUTIVE SUMMARY

Results-driven Software & AI Engineer specializing in high-performance web systems, creative WebGL/canvas engines, and full-stack architectures. Proven track record building autonomous financial analytics engines (Todar), adversarial AI systems (Red Gambit), and scalable educational platforms. Strong foundation in combinatorics, algorithmic problem solving, and low-latency interaction design.

### FEATURED TECHNICAL PROJECTS

#### Red Gambit - Adversarial AI Game Platform

<https://redgambit.lakshya.uk>

Tech Stack: [Next.js 14](#), [TypeScript](#), [Procedural Engine](#), [TailwindCSS](#)

- Architected a responsive adversarial strategy game platform featuring custom procedural audio synthesis and state-machine game logic.
- Engineered zero-latency Canvas/DOM hybrid rendering with sub-16ms frame times across desktop and mobile devices.
- Implemented modular telemetry tracking, custom UI state management, and real-time combat mechanics.

#### Todar 2.0 - Quantitative Financial Analysis & Intelligence

<https://todar.finance.lakshya.uk>

Tech Stack: [React](#), [Node.js](#), [Financial APIs](#), [D3 Data Visualization](#)

- Developed an institutional-grade stock and market intelligence terminal providing live ticker aggregation, fundamentals, and valuation metrics.
- Built responsive financial charting pipelines utilizing D3.js and WebGL for real-time order-book and historical price-trend rendering.
- Optimized data-fetching caching layers, reducing redundant API queries by 65% while maintaining live data freshness.

#### ElevateHub - Modern Collaborative Workflow Engine

Open Source

Tech Stack: [Next.js](#), [TailwindCSS](#), [Server Actions](#), [PostgreSQL](#)

- Built an enterprise task orchestration and workspace collaboration hub supporting real-time status synchronization.
- Designed an accessible, modular design token architecture with WCAG 2.2 AA compliant focus states and fluid micro-interactions.

### TECHNICAL ARSENAL

|                        |                                                                                                  |
|------------------------|--------------------------------------------------------------------------------------------------|
| Languages & Core:      | TypeScript, JavaScript (ESNext), Python, C++, HTML5/Canvas, SQL, Web Audio API                   |
| Frameworks & UI:       | Next.js 14 (App Router), React 18, TailwindCSS, Framer Motion, D3.js, Node.js                    |
| Tools & Architecture:  | Git/GitHub, Docker, Linux/Unix, REST APIs, GraphQL, Vite, Webpack, Vercel, Vitest, Playwright    |
| Engineering Practices: | Design Systems, WCAG 2.2 Accessibility, Core Web Vitals Optimization, State Machine Architecture |

### EDUCATION & VERIFIED CREDENTIALS

|                                                          |                                         |                   |
|----------------------------------------------------------|-----------------------------------------|-------------------|
| B.Tech in Computer Science and Engineering               | SRM Institute of Science and Technology | 2023 - Present    |
| Combinatorics and Probability (Specialization)           | UC San Diego Extended Studies           | Expected Apr 2026 |
| Korean Language & Intercultural Studies - Intermediate 1 | Sungkyunkwan University (Coursera)      | Mar 2025          |
| National Education Policy 2020 Certification             | SRMIST                                  | Feb 2025          |

### HONORS & LEADERSHIP

|                                                                                                                     |             |
|---------------------------------------------------------------------------------------------------------------------|-------------|
| - First Runner Up - Language Exchange SRM PBL 2024 (Represented Republic of Korea in international cultural summit) | 2024        |
| - Most Innovative Solution Award - UCSI Consulting Group, Malaysia (Business Model Strategy)                        | Jan 2025    |
| - Official EMCEE & Trainee Lead - SRMIST Directorate of Student Affairs & Directorate of Alumni Affairs             | 2024 - 2025 |